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Enhanced User Authorization Framework in Cloud Environments Using Weighted Proximity-Based Trust Evaluation

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ABSTRACT: Cloud computing is generally a service-based technology that offers technology services such as IaaS, PaaS, and SaaS through the internet. Since its inception, cloud computing has rapidly spread and is used in many web applications. The main advantage of these lies in the fact that the resources and services are accessible to thin clients. Although it seems advantageous, there are many vulnerabilities possible in terms of cyber threats and different attacks. Various levels of protection-based solutions are offered for cloud security, and access control is one of them. This paper presents a novel access control model for enhancing the security of the cloud. The proposed model uses the trust model for authorizing users to access different resources. The KNN model has recently been proposed for the prediction of trust, but the sensitivity in the present method for classifying the choices is not stable, especially when an imbalanced situation in the data arises. It is also noticed that the exponential distance, if chosen as the scheme for weighting, offers better classification performance and also lowers the variance. In this paper, a weights K-means nearest neighbor is proposed for the prediction of trust values of the user. The results have demonstrated that the proposed method is more efficient with respect to throughput and cost, and delay.

KEYWORDS: Access control-nearest neighbor, cloud environment, and trust value.

1 Introduction

Cloud computing is the brainchild of distributed computing that gives services to the stakeholders in different aspects. The objective of these systems is to give a scalable and online-demand-based service for the users at a low cost. The users have a lower overhead of installation of high-end computational devices. Also, the users do not need to procure the sophisticated infrastructure for high-end processing [1,2]. The important types of services that are offered through the cloud are Infrastructure as a Service, Platform as a Service, and Software as a Service. All these services are deployed in different models. There are four models in which these services are usually deployed, which are categorized as private, public, community, and cloud. One of the main reasons for organizations to move their business to the cloud is the ownership of data and processes. There may be a loss of control over the data and processes if used in a cloud environment. Further, these



are vulnerable to various attacks and threats. It is a common fact that both the organization and the cloud service provider must have trust in the users before starting the services. They should have a mutual trust exchanged between them before initiating the service. The users' trust in a CSP is based on the level of security and QoS. In turn, the CSP trusts the users based on the user profile and other parameters [3-5]. Recently, the KNN model has recently proposed for the prediction of trust, but the sensitivity in the present method for classifying the choices is not stable, especially when an imbalanced situation in data arises. It is also noticed that the exponential distance, if chosen as the scheme for weighting, offers better classification performance and also lowers the variance. Taking these factors, this paper proposes a k-nearest method for the prediction of the trust value of a user.

2 Related works

Jain et al [2016] introduced a trust method that used the QoS parameters for evaluating the trust. The fuzzy concept is used for the computation of the trust value of the service providers in the cloud systems. This offered an increased performance in the evaluation of the system. The parameters, such as the turnaround time, reliability, and availability, are considered for the experiment.

Tang et al [2017] introduced the trustworthy framework for the selection of a service provider called the TRUSS. The method aimed for the development of an enhanced trust-based evaluation of the QoS through an integrated approach for the trust evaluation using the combined objective as well as the subjective assessment of trust. The former is based on the monitoring of QoS, and the latter is based on the ratings through feedback. The experiments were performed using data which is synthesized, and the results show that the proposed method is better than the other state-of-the-art methods.

Shote et al [2019] focus on the security of data from the stakeholders' perspective, i.e., based on the ownership of data and the providers. Out of the analysis from the survey, it is found that there is still a high demand for a still more efficient method to be developed for the trust to secure the data, and also, user control policies need to be further investigated. The primary need is to improve the data access policies, and the second is to ensure QoS, and the third is to ensure that trust is maintained between both the users and the provider.

3 Proposed Methodology

The following section describes the proposed method in detail. The method uses a weighted k-nearest method for the prediction of trust for a given user. Figure 1 shows the proposed model framework. The framework consists of trust trust-dependent access control model that has different sub-module which uses ML techniques. If any user wants to get access to any cloud-based service, the request will be channeled through various levels of modules before authorization. This ensures the security of all the services and resources as well. The catalogue of the resource has different items with different reputations and trust values as well. When a certain user requests a particular resource, the framework checks for the trust of the user before granting access. If found as trustable, then the resources will be granted access, and this also ensures that there is a mutual trust maintained between the user and the CSP. The detailed description of each of the components is discussed below

3.1 Identity and Authentication Management (IDAM)

This is a module-based responsibility for the purpose of storing the user's authentication and information on their identity, such as the user credentials. It then acts as a perfect interface in-between the user and the cloud system. Activities such as the new user registration and update of any data of the users, and the removal of users are all taken care of by this module. Whenever a login request arises, the database of the identity is checked, and once there is a match, the user will be allowed to access the resource, and if not, the permission will be denied. Once authenticated, the

service request will be forwarded to the URM for initiating the level 2 authentication by this specific module.

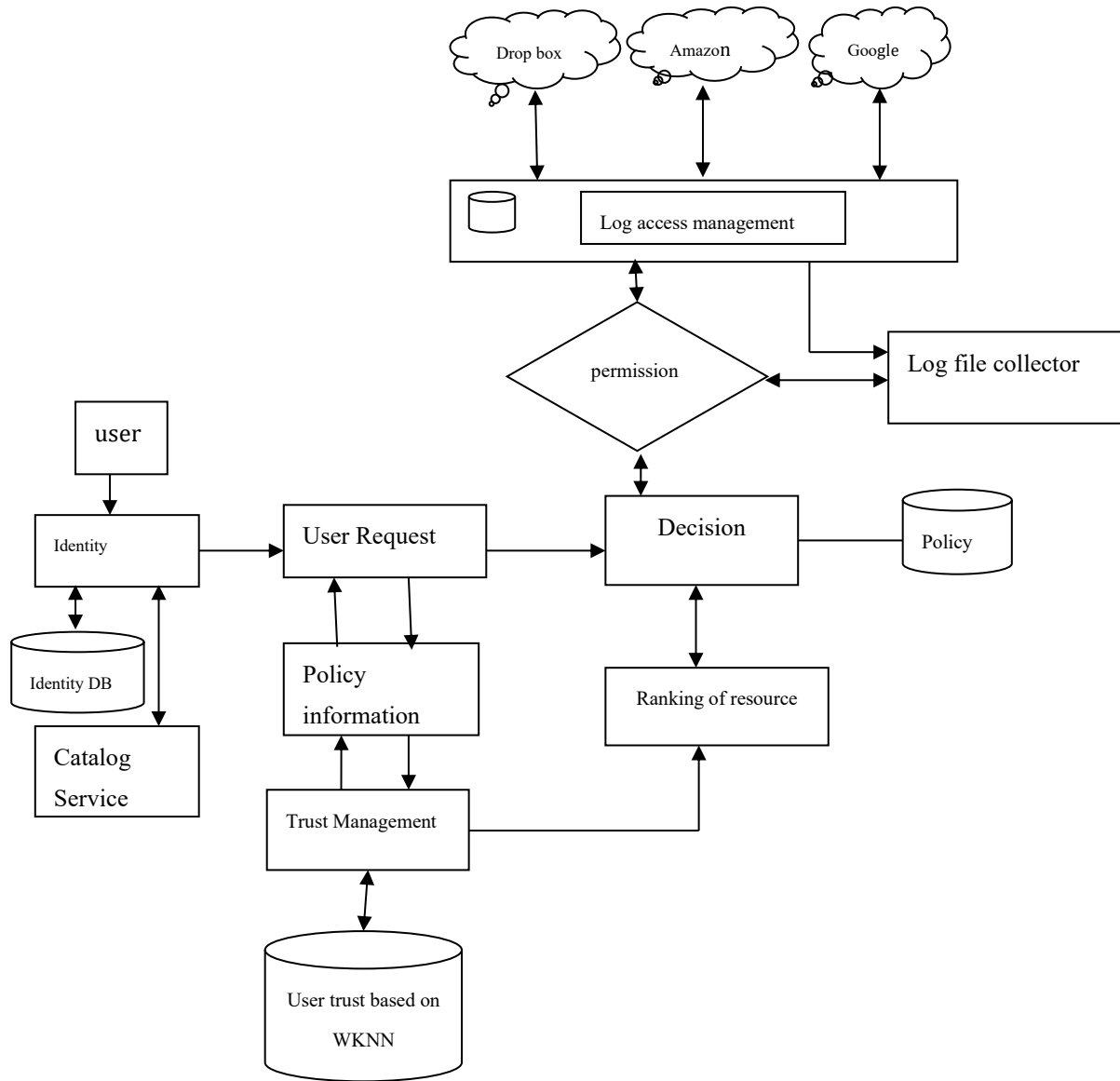


Figure 1: Overall Architecture of the Proposed Work

3.2 Catalog Service Management of Resource (CSMR):

The CSMR does the job of updating all the records pertaining to the services that are offered through cloud cloud-based service vendor (CSP). When a particular user enters the system, the requested service is looked up in the catalogue, and access will be granted based on the user's profile.

3.3 User Request Management (URM)

This particular module receives the user trust from the given PIP module, and then the same is forwarded as a request space vector toward making a decision and the policy enforcement module.

3.4 Policy Information point

The PIP module holds the responsibility of providing the users with a trust value when requests are received from a URM.

3.5 Decision and Policy Enforcement

The DPE module takes a vital role in the decision-making regarding whether a particular resource which is requested to be allotted or not. As and when the request is received by a DPE, the URM fetches the corresponding policy from the database. It then performs the comparison of all the details of the users with that of the given policies and the threshold value, and then appropriate access would be either granted or denied. The log file module collects all the information pertaining to the user, whether valid or invalid. Through this, a mutual trust is maintained between the organization and CSP.

3.6 Log Access Management (LAM)

This module is responsible for the secure access of the users to the services of the cloud. The activities and the corresponding behavior of the users are kept stored in logs. Once the execution of a request is completed, feedback will be submitted back to the trust module. As this maintains a log of trusts of the users, these can be used in the future to update the same to be updated with new trust values.

3.7 Ranking of Resources Management

The RRM module has the complete list of resources that are available in the cloud. All these resources are then ranked according to the capability and the performance that can be updated regularly when the RRM gets a trusted reputation from the cloud. There are several trust-based methods for controlling access that use ML techniques. The decision of which better resource to be allotted to a user is based on the decision taken by RRM.

3.8 Trust Management Module (TTM)

The TTM takes the responsibility of a real-time assignment of the value of the user's trust based on evidence. These evidences are obtained from the user behavior and is used for the evaluation of trust. This module takes the log file that projects all the interactions of a user that are based on the cloud, and it recomputes the trust value. It also computes the reputation that a particular resource has on the cloud and sends the trust value back to RRM for assigning a rank to the trust.

3.9 Authorization Process

Whenever a user wishes to access any of the cloud services, a request is required to be submitted to the CSP. The services are, in turn, provided by the cloud without any negotiation by the CSP. The negotiations would have already been covered in the service level agreement (SLA). The following section describes the process of authentication step by step.

1. Only after the successful authentication by IDAM are the users allowed to request a particular resource in the cloud.
2. The users are enabled with the option to choose the services required from the available catalogue maintained by the CSMR. The module then redirects the same to the IDAM.
3. The IDAM, in turn, collects the requirements of the user from the CSMR, and further, a request is framed through the request information vector to the URM.
4. The URM sends the user identity to the PIP module for the evaluation of trust value, and the PIP forwards the same to the TMM. The TMM at last forwards the trust to the PIP.
5. Now, the URM needs to add the value of the trust of a given user in terms of a vector, and then forward the decision and the policy module (DPE).

6. The DPE then obtains information from the URM. The DPE then makes a comparison between the vector that is received with the needed policies for accessing the database of policies.

7. The RRM module then selects the most trusted and suitable resource as per the job and then sends it to the DPE. It enables secure access for the user to get the requested resource. It also has the logs that are forwarded to the TMM.

8. The TMM finally gives secure access for the user, such that the user can access the service in a safe mode. It also has log files, which are forwarded later. The TMM keeps processing the log files of a user and their behavior.

The proposed model was able to achieve the secured accesses that are based on a mutual level of trust.

3.10 Trust Management Module (TMM)

The TMM takes the responsibility of a real-time assignment of the value of the user's trust based on evidence. These evidences are obtained from the user behavior and is used for the evaluation of trust. This module takes the log file that projects all the interactions of a user that are based on the cloud, and it re-computes the trust value. It also computes the reputation that a particular resource has on the cloud and sends the trust value back to RRM for assigning a rank to the trust. The model is depicted in Figure 2. TMM is not biased towards any of the service providers or the users. It can also be noticed that the TMM has different phases for computing the trust. The following section describes how the value of trust can be calculated based on the various functionalities.

3.10.1 Cloud user

Each user needs to register prior to getting access to the resources. The database maintained for the identity has all the information about the users. When a particular user requests a resource, the authenticity and the authorization are checked through different components, and the services will be accessible only if the process is successful.

3.10.2 Resources

Both the hardware and the software can be a resource. All the cloud-based resources are prioritized based on the pre-defined reputation and the trust value of the cloud. The trust-based access controls use different ML techniques before providing the grant to the particular resource.

3.10.3 Log file of user behavior

The logs that are created when the LAM module records all the activities that are performed by the user in a given cloud. The same will capture the realistic behavior of all users through the trust that is predicted using ML methods.

3.10.4 SLA monitoring module

This module captures the real-time allocation of services in the cloud. The performances of the resources are also monitored in this module.

3.10.5 Feature extraction module

This module feeds in all the logs. The module tends to capture all the relevant parameters and the respective values from the log files. These values are then passed to the ML-based training algorithms and then to the predictive module. In the proposed method, the log files from the Apache servers are considered. Each of the parameters is then combined with the weighted factors, and the corresponding strengths are determined by the factor of contribution. Here, some of the parameters are found to be more dominant than the others. The parameters $P_1, P_2, \dots, P_n$ is intended to detect attacks and other unauthorized access requests. The user's behavior analysis is done, and the corresponding output is to be included in calculating the trust values.

3.11 Trust Value Prediction Using Weighted K-Nearest Neighbor

The module takes up parameters and data from the FE module. The data is then fed to the corresponding K-means model, and training is done. In the conventional KNN model, the classification sensitivity towards the choices of neighbors exists particularly in the imbalanced data. Hence, we introduced the weighting schemes for better classification and accurate performance, and also low variance.

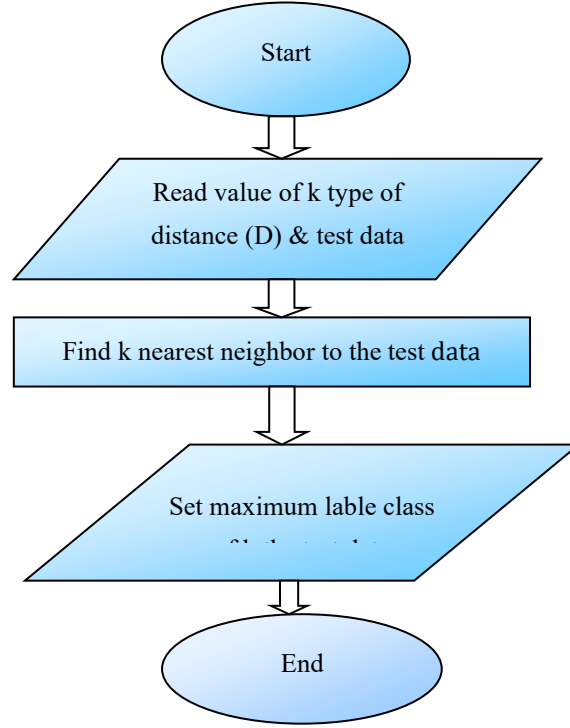


Figure 2: Flow chart for KNN

The proposed method takes advantage of the distance calculation for the purpose of classification, with the belief that there must be a concrete candidate that is chosen for the scheme of weighting. In this proposed method, an improved KNN is introduced to conquer the influence of the neighbors of K in case of precipitation in the data. Given a training set $T = \{\{x_n \in R^d\}_{n=1}^N\}$ that has M number of classes and N denotes the random numbers of the set T, and d represents the dimension of the feature. The labels of the class are obtained through the query point x are obtained through the steps as follows [9-11].

First, the K nearest values for the query points that are not known in a given set T are identified, and let $\bar{T} = \{x_i^{NN}, c_i^{NN}\}_{i=1}^k$ represent the k-near value of x, and let K near neighbor are denoted by denote the set of $x_1^{NN}, x_2^{NN}, \dots, x_k^{NN}$ are then sorted in ascending order based on the respective Euclidean distance to x as described in Eq(1). Allocation of various weights to the K near neighbors, and the weight of the X and J-1 neighbor is calculated as

$$\omega_j = \begin{cases} \exp\left(-\left(\frac{d_k - d_j}{d_k - d_1}\right) \cdot \left(\frac{d_k + d_j}{d_k + d_1}\right)\right), & d_k \neq d_1 \\ 1, & d_k = d_1 \end{cases} \quad (1)$$

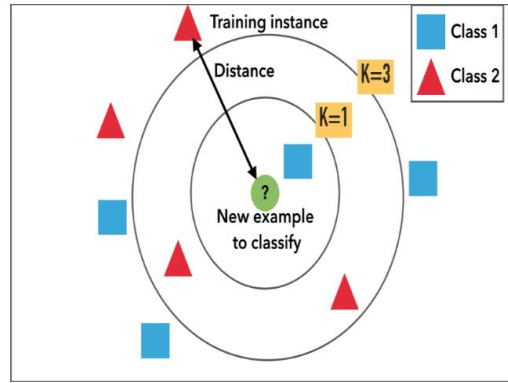


Figure 3: Working procedure of k-nearest neighbor

Classification of the query point x of the class C is done by the majority weighted voting of the neighbors.

$$\bar{c} = \underset{c}{\arg \max} \sum_{(x_j^{NN}, c_j^{NN}) \in \bar{T}} \omega_j \cdot (c = c_j^{NN}) \quad (2)$$

WEIGHTED K-NEAREST NEIGHBOR

The steps are given as under

1. Calculate the distances of the nearest neighbors of the query
2. Arrange the distances in increasing order
3. Search the k-nearest neighbors of the query X -
4. Compute the weight of the K-Nearest Neighbors
5. For $i=1$ to K , do Steps 1-5
6. End for
7. Assign the major weighted voter class label $Y1$ and $X1$
8. End

4 Results and Discussion

Owing to the wide range of the cloud and the practical difficulty in conducting real-time experiments, the JAVA simulator and the cloud-sim are used for experimental purposes. In order to evaluate the proposed WKNN model, the comparison is made with that of the other methods, taking the performance metrics as Throughput, Delay, and Accuracy, and the benchmark methods such as FL and KNN. Figure 4 depicts the performance outcomes for the existing FL, KNN, and the proposed technique as of throughput is considered. From the figure, it seems that the proposed WKNN model has more throughput than the FL and KNN methods, with 590 kbps.

Table 1: Performance comparison results

Metrics	Methods		
	FL	KNN	WKNN
Throughput(kbps)	450(kbps)	572(kbps)	590(kbps)
Delay(sec)	700(sec)	650(sec)	530(sec)

Accuracy (%)	74	82	90
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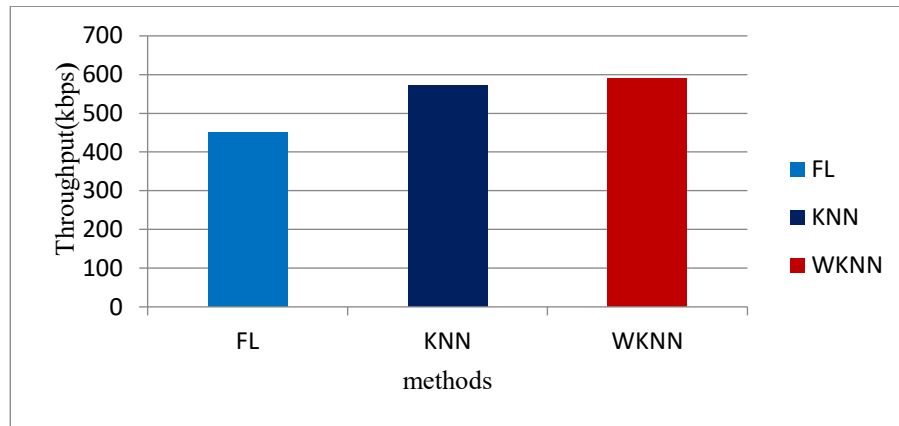


Figure 4: Throughput results vs. classification methods

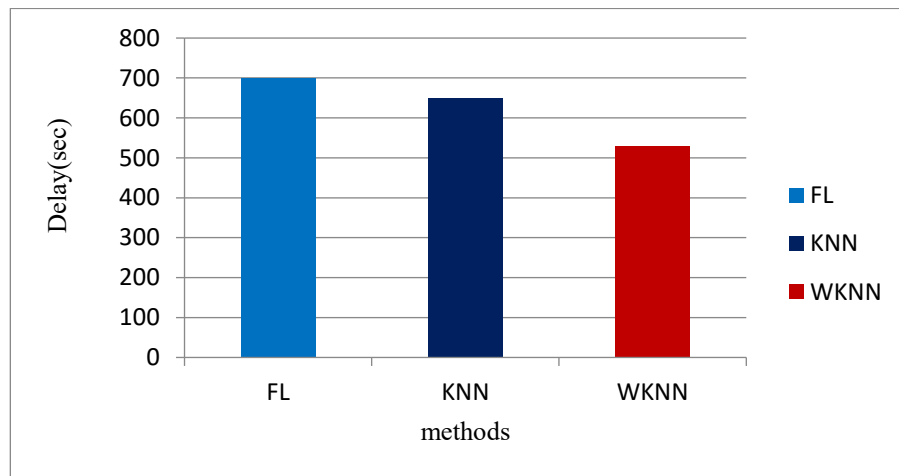


Figure 5: Delay results vs. classification methods

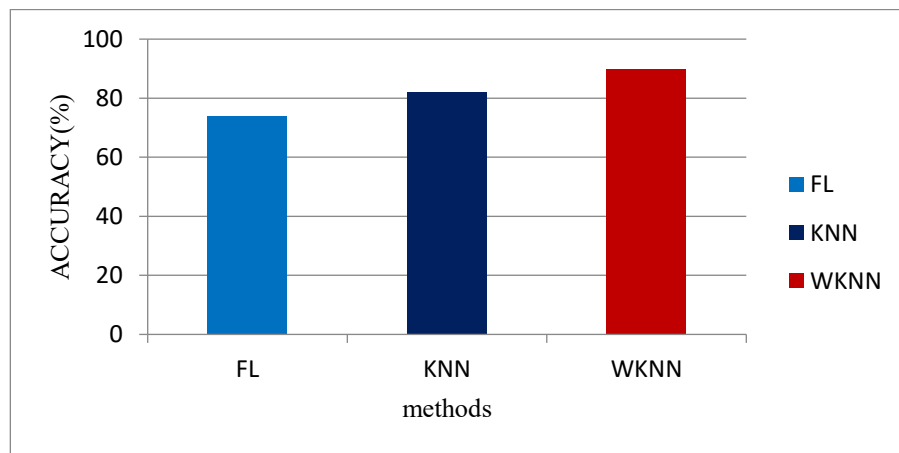


Figure 6: Accuracy results vs. classification methods

Figure 5 shows the graph, which is the comparative analysis of the FL and KNN with the proposed method, taking the delay as the metric. From the graph, it is evident that the proposed model has a lower delay of 530 seconds when compared to FL (700S) and KNN (650S), respectively.

Figure 6 shows the graph, which is the comparative analysis of the FL and KNN with the proposed method, taking the accuracy as the metric. From the graph, it is evident that the proposed model has a high accuracy of 87% when compared to FL (68%) and KNN (81%), respectively.

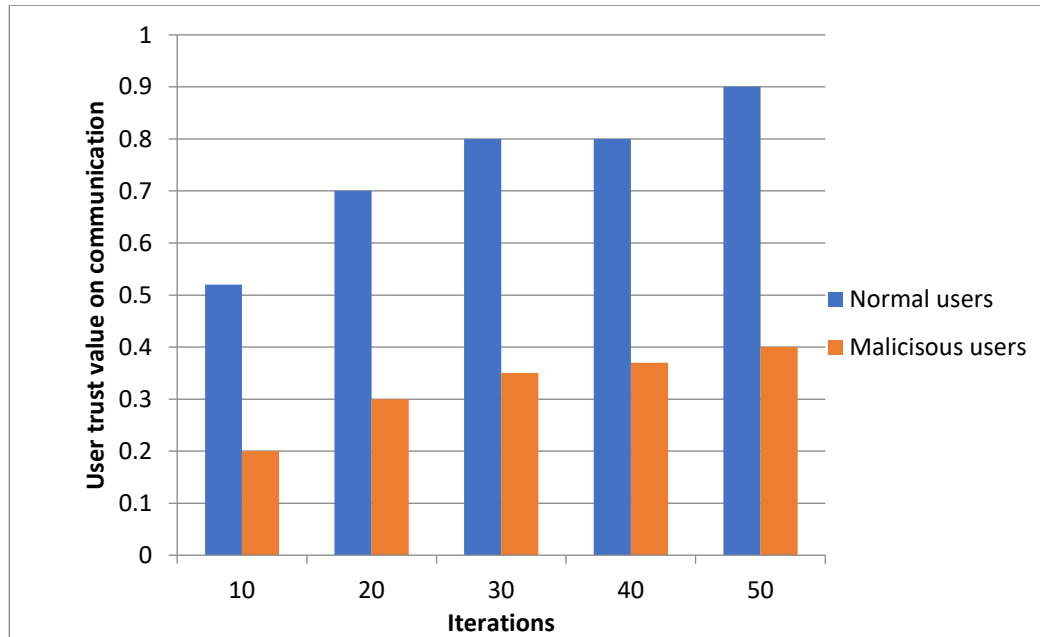


Figure 7: User trust value on communication results vs. iterations

Figure 7 depicts the trust value of the communication outcomes in the proposed WKNN technique. From the graph, it is seen that the X-axis refers to the iteration and the Y-axis refers to the value of trust in the communication. From the graph, it is seen that the malicious users who have a trust value below 0.5 are detected by the proposed method.

5 Conclusion and Future Work

This paper presents an effective evaluation mechanism for trust using the weighted K nearest neighbor for accurate prediction of trust of a user using the weighted K nearest neighbor algorithm. It is identified that the weighted scheme shows more accuracy and better classification performance with less variance that the user to be trusted before accessing the resource. The contributions of the proposed system are pointed out as follows: 1. It enables a scheme of trust evaluation of the efficiency of the cloud 2. It aimed to ease the IaaS selection process and offer more stability in the service when using clouds through the consideration of performance in the selection phase. The future research is aimed at building different policies for selection for cloud usage and brokers based on the proposed method to further automate the deployment.

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Conflicts of Interest: The authors declare no conflicts of interest to report regarding the present study.

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